DraftFM: Zero-Shot Drafting of Unseen *Magic: The Gathering* Sets from Public Card Features

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Abstract

Draft assistants for *Magic: The Gathering* require weeks of post-release human pick data before they can model a new set; on release day they have nothing. We train DraftFM, a pick model that represents every card as a frozen vector of public information (structured Scryfall features plus a sentence embedding of rules text) and carries no set-identity parameters, so it can score a set the moment its card list is public. The 2.2M-parameter recipe used for architecture and hyperparameter selection, trained on 149.4M human picks from 28 sets of 17Lands public data, reaches 54.3% expert-pick agreement zero-shot on three held-out sets (48.9% / 57.5% / 56.5% on BRO / TMT / SOS), 78.6% of the accuracy of per-set models trained directly on each target set by the pre-registered normalized score (identical held-out picks), and above every day-one-feasible published number we could locate, each measured on a different test set than DraftFM's, so these situate rather than compare head-to-head (Table 2). A pre-registered ablation on the same held-out trio found a smaller recipe that is no worse within seed noise: dropping the set-context tower cuts the model to 1.6 M parameters, with zero-shot agreement of 54.8% against 54.3% for the with-tower recipe – a gap smaller than the 0.5pp seed spread measured on this architecture (§7). That tower-free recipe is F-full, the model whose one pre-registered inference pass over the first public snapshot of MSH – a licensed-IP set whose pick data nothing in the pipeline has touched – is the headline evaluation ([PENDING: F-full MSH top-1]% expert top-1). We release the code, the frozen protocol, model weights, and per-pick prediction files.

54.3%
zero-shot expert-pick agreement, dev-selection recipe (§6)

78.6%
of a per-set-trained ceiling's accuracy (pre-registered normalized score), with zero picks from the target set

1.6M params
the shipped recipe (A-noctx architecture); F-full trains from scratch in 2.1 h on one workstation (§4.3)

1 Introduction

Drafting in *Magic: The Gathering* is a sequential game of imperfect information: eight players pass packs of cards, each privately assembling a deck one pick at a time. Strong pick models exist.

Supervised models trained on human draft logs reach 48.7–71% agreement with human picks within a set [5, 6, 10] (a general, not expert-filtered, pick population; experts are easier to predict, scoring several points higher), but they are set-specific. Each model encodes cards as vocabulary indices, learns from picks made in that set, and transfers nowhere. Because public draft logs appear on a delay, a data-driven assistant for a new set cannot exist until roughly two weeks after release [5]. On release day, the state of the art is a spreadsheet of hand-tuned expert ratings [7].

This paper asks how far pure transfer can go. DraftFM is a single pick model trained across every public 17Lands draft set, in which a card is nothing but a frozen vector of public information: structured features parsed from its Scryfall record and a sentence embedding of its rules text. The model has no set-identity parameters and no per-card weights. Whatever it knows about a set it must infer from the cards in the pack and the drafter's accumulating pool, not from a memorized identity. That inference transfers to a set released tomorrow; a memorized card vocabulary cannot.

The evaluation is designed against a failure mode endemic to this setting: a "zero-shot" number quietly tuned on its own test set. Our protocol (§5) was frozen, in a tagged public commit, before the test set existed. The test set is MSH, a licensed-IP set whose public draft data had not been published at the freeze; no training, tuning, or metric selection touches MSH pick data, and the headline number is produced by a single pre-registered inference pass over the first public snapshot. All model selection happens on three development sets (BRO, TMT, SOS) held out of training, chosen so our numbers are directly comparable to the only published multi-set feature-based zero-shot benchmark [3] (same BRO holdout) and to our own strongest per-set ceilings.

Zero-shot on the development trio, DraftFM agrees with expert picks 54.3% of the time (48.9% / 57.5% / 56.5% on BRO / TMT / SOS): 78.6% of what per-set models trained on each target achieve by the pre-registered normalized score (identical held-out picks, §5), and above every published number attainable on release day that we could locate (expert-tuned ratings 44.5% [10] on M19, an hour-zero rarity heuristic 42.8%, measured on SOS as a rehearsal stand-in rather than per dev set) as well as zero-shot GPT-4o (43% [2] on NEO) – each on a different test set than DraftFM's, so these situate rather than compare head-to-head (Table 2); the one same-set comparison available is the rarity heuristic against DraftFM's own SOS number (56.5%). On MSH the frozen evaluation gives [PENDING: F-full MSH top-1]% expert top-1 agreement.

Contributions:

- **A cross-set model and recipe.** A two-tower architecture over frozen card features, refined by a pre-registered ablation from a 2.2M-parameter design with a set-context tower to the 1.6M-parameter tower-free recipe that ships (A-noctx, §7), training on 149.4M picks in about 1.8 hours on a single consumer workstation (§4).
- **A frozen zero-shot benchmark.** A pre-registered, tamper-evident protocol for day-one draft evaluation, plus the first multi-set zero-shot numbers that use no post-release information of any kind (§5, §6).
- **A scaling study.** Zero-shot accuracy as a function of training-set count from 1 to 28 sets, with set-composition probes and zero-shot seed bands at two rungs (§6; the bands bound the curve's seed noise rather than fully characterize it, see caveat there).
- **A licensed-IP shift analysis.** Pre-registered ablations designed to isolate what text embeddings contribute when a set's names and flavor come from a licensed universe rather than in-universe *Magic: The Gathering* (§7).
- **Release.** Code, the frozen protocol, data manifests, model weights, and per-pick prediction files for every number in this paper (Appendix B).

Figure 1: Emeritus of Ideation, an actual first pick from a real thirteen-card SOS pack (empty pool). The per-set ceiling model (Table 1) ranks it above every other card in the pack by a wide margin: real expected value[1] 8.61, seen in 65.1% of games it's in, average last-seen-at pick 1.4, and >99% model dominance over the closest candidate in the pack (Quandrix Charm). SOS is a public development set, never the frozen MSH test set (§5); this figure illustrates the model's output, not a headline result. Numbers regenerated by `figures/make_verdict_panels.py`.

2 Related work

Generalised card representations. Bertram, Fürnkranz, and Müller [3] established the zero-shot drafting problem, and their paper remains the only published quantitative result on it from multi-set feature-based training (a separate zero-shot LLM number exists, discussed below under "LLMs as drafters"). They replace one-hot card identities with concatenated feature vectors: hand-engineered attributes plus a Sentence-BERT embedding [8] of card text (1306-d), a learned autoencoding of card images (1024-d), and 16 post-release usage statistics from 17Lands. This lives inside a contextual-preference-ranking model trained with a triplet loss. Two of their findings shape our design: feature representations match one-hot accuracy within a set (nothing is lost by generalising), and multi-set pretraining, not the representation alone, is what buys transfer. Pretrained on 13 sets and evaluated on held-out BRO, their best representation reaches 55.4% pick agreement. That representation includes the usage-statistics block, which is computed from human play after a set's release; the number is therefore an anchor for zero-shot *card generalisation* rather than for day-one deployment, where such statistics cannot exist. The same paper's day-one-feasible representation (features only) reaches 33.6–35.6% on unseen sets under single-set training. No multi-set features-only number is published. Filling that gap (multi-set training, public card information only) is this paper's headline experiment, and we adopt their held-out BRO as a development set so the comparison is on the same holdout. A companion paper [4] improves the within-set triplet objective with an adapted InfoNCE loss but reports no unseen-set results.

Within-set pick models. Ward et al. [10] compare heuristic and neural drafters on M19 and report the two most useful reference points below the data-driven regime: an expert-tuned rating bot at 44.5% top-1 agreement and a random agent at 22%. Their neural drafter (48.7%) and

[1] *Expected value* is this per-set model's own raw score for the card (higher is better; not a model input, and not the top-1 agreement metric reported elsewhere in this paper). *GIH%* and *ALSA* are 17Lands per-card statistics reported for context: GIH% is win rate in games where the card was drawn, and ALSA is the average pick number at which the card was last seen (lower means drafters value it more highly). *Model dominance* is this paper's own pairwise measure, sigmoid(top EV – runner-up EV).

its successors, the statistical-drafting MLPs deployed at statisticaldrafting.com [5] (∼70%) and Czerner's puder transformer [6] (71%), define the within-set state of the art, and the statistical-drafting architecture is the direct lineage of the per-set ceiling models we compare against. All are one-hot models over a fixed card vocabulary; none can score an unseen set. The deployment gap is explicit: statistical-drafting documents that no model can ship until 17Lands data appears, roughly two weeks post-release [5], and Draftsim's day-one ratings are written by human experts [7].

LLMs as drafters. UrzaGPT [2] evaluates large language models on draft picks. Zero-shot GPT-4o reaches 43%, narrowly above the hour-zero rarity heuristic we measure (§6). We still mark it non-day-one in Table 2: NEO shipped years before GPT-4o's training cutoff, so its pretraining corpus cannot be shown free of the set's own post-release strategy discussion, a leak a frozen feature-only encoder cannot have. Providing full rules text in the prompt lowers GPT-4o's accuracy further. Open 7–8B models fail to pick legal cards until LoRA-tuned on a million picks from the target set, after which they still trail small supervised MLPs at orders of magnitude more inference cost. We take the implied guidance: use a frozen text *embedding* as one feature block, not a language model as the policy.

Text-grounded transfer in card games. Cardsformer [11] encodes Hearthstone card descriptions with a language model so a gameplay policy generalises to unseen cards. It is the closest cross-game precedent for text-mediated transfer, applied to gameplay rather than drafting.

3 Data

Corpus. Training data is the complete 17Lands public draft-data corpus [1], spanning every set the model has to generalise across: 31 expansions from STX (April 2021) through SOS (June 2026), Premier (best-of-one) and Traditional (best-of-three) draft where published, 59 (set, format) files, roughly 160M pick rows. Each row is one human pick: the pack contents, the drafter's pool, the card taken, and two coarse skill covariates (recent game-count bucket and win-rate bucket). Files were fetched once from the public S3 bucket, per the 17Lands usage guidelines (bulk downloads only), and frozen by sha256 and ETag in a data manifest (content hash `443bd25f72ee`); the manifest is released with the paper.

Two public files are structurally excluded from the corpus, recorded in the corpus registry: OM1 (a pick-two format whose rows carry two picks and whose pool columns undercount them, and whose digital-only cards largely do not exist on Scryfall) and the Powered Cube (a curated 545-card singleton pool with no expansion structure). A third, the VOW QuickDraft file (human picks inside bot pods, off-distribution pack dynamics), is held out of the default corpus for a distributional reason rather than a structural one: it is one of the two pre-registered flexibility-clause extras (§5) that may join F-full's training data if a dev-trio ablation shows a gain. Everything else is kept, including digital-only and remaster sets.

Normalization. The corpus spans three schema eras and several quirks, all normalised at ETL time: the 2021 files are gzipped tarballs rather than plain gzipped CSVs, and the oldest era reports match-based rather than game-based skill buckets (harmonised; identical in best-of-one). Five sets are missing the first pick of each draft in the source logs (an annotation, not an exclusion), and packs are 13, 14, or 15 picks depending on the set (a model input, §4). Card names are joined to Scryfall by normalised name with an explicit alias table; unresolved names fail the build rather than featurizing as zeros. Per set we retain every valid pick and tag it with the drafter's skill covariates: training filters no one out. Skill is a conditioning input ("decision + tag by venue").

Splits. Within each (set, format) shard, drafts are split by a hash of the draft identifier (`crc32 % 1000 < 50`, $\sim$5% validation). Early stopping uses only this within-training-set validation split. The *expert slice* used for all headline evaluation is picks by drafters with win-rate bucket ≥ 0.55 and game-count bucket ≥ 100, the same filter the per-set ceiling models train and evaluate on, so ceiling comparisons are population-matched.

The held-out sets. Four sets never contribute a pick to model selection. BRO, TMT, and SOS are the development trio (§5): all model selection keys on zero-shot accuracy over these three, computed from *F-dev*, which trains on none of them. Once selection is frozen, the headline model (*F-full*) retrains on all 31 sets, dev trio included: only MSH stays unseen by every model in the battery. MSH, the test set, is stricter still: a registry-level gate (`EVAL_ONLY`) makes every training and feature-fitting code path refuse MSH pick data outright, and at the protocol freeze its public data did not yet exist. MSH *card* features from Scryfall are permitted. That is the zero-shot contract: public card information only, no human behavioral data of any kind. "Unseen" describes the set, not every card in it: BRO/TMT/SOS packs include some ordinary reprints (mostly base-set staples and basic lands) that also appear in F-dev's training corpus under other sets. This confers no advantage under the zero-shot contract above – a reprint has no memorized identity to exploit, only the same frozen feature vector any novel card with identical features would get – but it does mean set-level and card-level novelty are not the same claim.

4 Method

4.1 Cards as frozen feature vectors

Every card is a fixed 775-dimensional vector computed from public information alone, frozen before training.

Structured features (391-d). Parsed from the card's Scryfall record: mana value (scaled + bucketed one-hot, 10), mana-cost pips including hybrid/Phyrexian/X structure (10), colors (8), color identity (5), supertypes (3), card types (9), creature subtypes (top-128 vocabulary + an unmatched count, 129), power / toughness / loyalty each as (scaled, missing, star) triples (9), rarity (5), keyword abilities (166 slots + unmatched count, 167), layout including back-face flags for double-faced cards (10), and text-shape features: length, line count, and 24 regex flags over rules text for common functional categories (removal, card draw, counterspells, tokens, ramp, . . .) (26). Double-faced and split cards take numeric blocks from the front face. Subtype and keyword vocabularies are counted over *training* sets only; a registry gate refuses to fit vocabularies on MSH, the frozen test set (§5). This blindness is guaranteed for MSH specifically; it does not mean the dev trio's cards were excluded from every vocabulary in the release (§3), so "dev-trio zero-shot" is a claim about training picks, not about vocabulary provenance. The resulting featurizer manifest is content-hashed (`e1313cadd03b`) and released; held-out cards featurize through the frozen manifest, with out-of-vocabulary subtypes and keywords absorbed by the unmatched counters. Scalars are clipped to $[0, 1]$; parse failures set a missing indicator, never a silent zero.

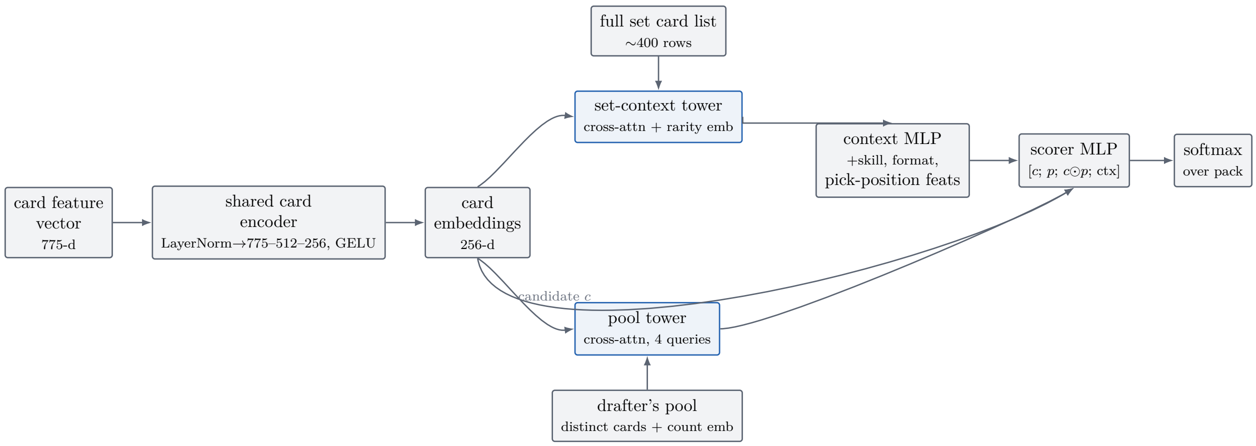

Figure 2: The two-tower scorer (§4.2). A shared card encoder embeds every card once per (set, format) step; a pool tower summarises the drafter's held cards, a set-context tower summarises the full card list, and a per-candidate scorer combines the candidate embedding c, the pool summary p, their elementwise product, and the context vector into a softmax over the pack.

Text embedding (384-d). The card's type line and rules text, embedded with a frozen `bge-small-en-v1.5` sentence encoder [12] (L_2-normalised). The card's own name (and, for legendary cards, its short name) is masked to a placeholder token before embedding, so the encoder cannot key on names it may know from pretraining, a deliberate handicap that matters for licensed-IP sets (§7). Parenthetical reminder text is kept: for a brand-new mechanic it is the only definition the model will ever see. Back faces are appended for double-faced cards.

4.2 Architecture

DraftFM is a two-tower scorer with 2.2M parameters (Figure 2). Its two design invariants operationalize the zero-shot contract (§3):

1. **No set identity.** There is no set embedding and no per-card parameter anywhere. The only way the model can know what set it is drafting is a *set-context tower* that reads the set's full card list, plus the drafter's accumulating pool. "Blue is the aggressive color in this set" must be inferred from the card list, not memorized as a set identifier. Whether the tower itself turns out to be necessary for that inference, or whether the invariant holds just as well without it, is addressed later in this section.

2. **Skill conditions the scorer, not the cards.** Card semantics are skill-invariant; the pick policy is not. Skill-bucket embeddings enter only the context pathway. Serving conditions on a fixed high-skill bucket; training sees every pick tagged with its drafter's bucket.

A shared *card encoder* (LayerNorm → 775–512–256 MLP, GELU) maps feature vectors to d=256 embeddings. Because training batches are homogeneous in (set, format), the encoder runs once per step over the set's full card list (∼400 rows) and everything downstream gathers rows from that table. The *pool tower* represents the drafter's pool as the set of distinct cards held, each embedding translated by a learned count embedding (counts capped at 8), pooled by cross-attention from four learned queries (4 heads); an empty pool attends over a learned null token. The *set-context tower* applies the same query-pooling to the entire card list (with a rarity embedding added) to produce the set summary. A context MLP combines the set summary with skill and format embeddings, seven pick-position features (pack/pick indices, pool size), and four set-shape scalars (card-list size, picks-per-pack indicator) into a 64-d context. Each candidate c in the pack is scored by an MLP over $[\,c;\ p;\ c \odot p;\ \mathrm{ctx}\,]$, where p is the pool summary and $c \odot p$ the elementwise interaction; a softmax over the pack's real slots gives the pick distribution.

The set-context tower's contribution is tested directly: the A-noctx ablation (Table 4, §7) removes it and scores higher on all three dev sets, and that tower-free configuration – not the one just described – is what F-full and the frozen MSH evaluation run. The first invariant holds either way, since no set-identity parameter exists in either configuration; dropping the tower only means the shipped model's sole route to set-relative signal is the pool tower, each candidate's own features, and the four set-shape scalars above, never a read of the rest of the card list.

4.3 Training

One recipe for every run in the paper (fixed before the scaling and ablation studies, since the protocol forbids per-rung tuning). Batches of 8,192 picks are drawn from one (set, format) shard at a time, shards sampled proportionally to $n_{\mathrm{picks}}^{0.5}$, an interpolation between pick-uniform (which lets the largest sets dominate) and set-uniform (which oversamples tiny sets). AdamW (β_1=0.9, β_2=0.98, weight decay 0.01), peak learning rate 10^{-3}, 2,000-step linear warmup, cosine decay; cross-entropy with label smoothing 0.05 renormalised over real pack slots. The presentation budget is four epochs. Within-training validation (never the dev sets) is evaluated every 2,000 steps and training stops after three non-improving evaluations. Seed 17 throughout. Seed variance is measured at two scaling rungs (§5).

Training runs on a single Apple M3 Ultra workstation (96 GB unified memory) in PyTorch 2.12 eager mode on the MPS backend, at ∼42,750 picks/s. The full 149.4M-pick run takes about 1.8 hours, stopping at step 34,000. Training-backend correctness is not assumed and is actively gated rather than hoped for; specifics are in the companion reference. Inference exports to ONNX (opset 18) and runs on a consumer x86-64 machine, CPU-only, well under a millisecond per pick.

5 Pre-registered evaluation protocol

The evaluation is frozen at a public git tag (`eval-protocol-v1`) that predates the existence of the test data. The tag fixes the protocol document and the metric implementations, with a hash guard that refuses to run if the metric module drifts from the tagged version. The model battery is anchored separately, by the sha256 of each artifact recorded in the experiment ledger before the test data was downloaded; the battery file itself is not part of the tagged tree. There is one evaluation round. Any second round would require a new public tag and disclosure of the total number of rounds.

5.1 Test set: MSH, exactly once

MSH is the first *Magic: The Gathering* set drafted on Arena after the protocol freeze: a licensed-IP (*Universes Beyond*) set, which makes the test deliberately harder (§7). The test snapshot is the first public 17Lands MSH Premier file successfully downloaded on or after its publication, frozen immediately by sha256 and ETag and never re-downloaded. Until the frozen evaluation has run, no training, tuning, metric-peeking, or feature-vocabulary fitting may read MSH pick data. MSH *card* features alone are legal (the zero-shot contract, §3). Quality gates at snapshot time: modern schema, Premier rows present, at least 2,500 expert-slice drafts, $\geq$ 99% of pack card names joining to card features. On a volume failure the single pre-declared contingency is to wait exactly one snapshot cycle. The evaluation executes within 72 hours of the gates passing: one inference pass per battery member over the frozen snapshot, cached as per-pick prediction files from which every statistic is computed. Models are never re-run during analysis.

Two populations are scored. The *expert slice* (§3) is the headline; the all-users slice is secondary. Two conditioning configurations are pre-registered: *deployment mode* (headline), conditioning on a fixed top skill bucket selected on the dev sets and frozen into the battery configuration; and *human-model mode*, conditioning on each drafter's true bucket, scored on all users. Forced picks (final pick of a pack) are included in headline numbers, matching the per-set ceilings and prior work; non-forced variants are secondary rows.

5.2 Development discipline

Every architecture, hyperparameter, text-encoder, and conditioning decision keys on one number: the unweighted mean over {BRO, TMT, SOS} of expert-slice deployment-mode top-1 (ties broken by mean log-loss), computed zero-shot from a single development run (*F-dev*) trained on the universe minus those three sets and MSH. The trio is chosen for external validity: BRO is the same holdout as the only published multi-set feature-based zero-shot number [3], SOS carries our most mature per-set ceiling, and TMT is a licensed-IP set: the rehearsal for MSH's distribution shift. Early stopping never sees dev metrics (§4.3), so the scaling curve is not selected on its own y-axis.

5.3 Metrics and statistics

Primary: top-1 agreement with the human pick, expert slice, deployment mode. Secondary: top-3; all-users top-1 (human-model mode); the *normalized score*—zero-shot top-1 divided by the per-set ceiling's top-1 on identical picks (aligned pick-by-pick on the ceiling's validation split); mean log-loss; top-label expected calibration error (15 equal-mass bins; a temperature is fit on the dev trio only and frozen into the battery before T0, never fit on the test set itself, §7); per-(pack, pick) accuracy curves against the per-cell random floor; and *late-draft retention*, the model/ceiling ratio restricted to picks 8+ of each pack, where pool context dominates.

All intervals are cluster bootstraps over drafts, not picks (B=2,000, fixed seed, percentile 95% CIs); paired comparisons share resample indices. Clustering matters little but is grounded empirically: measured intraclass correlation of pick correctness within drafts is $\rho = 0.0099$ (per-set SOS validation; 1,662 drafts, 69,803 picks), a design effect of $\sim$1.4 at $\sim$42 picks per draft, giving CI half-widths $\sim \pm\, 0.36$pp at the 2,500-expert-draft gate (binomial approximation at p=0.5). Seed variance is measured with three seeds at two scaling rungs, and set-composition variance with two probe rungs (§6).

5.4 The battery

Frozen by artifact hash before the test snapshot: (1) **F-full**, the final recipe trained on all 31 sets, the headline model; its dev-set numbers are meaningless (it trains on them) and are never quoted. A pre-registered flexibility clause lets two optional extras (the Powered Cube file and VOW QuickDraft) join F-full's training data if and only if a dev-trio ablation shows a gain; the decision is frozen with the battery and reported either way. (2) **F-dev**, the development model (§above), which doubles as the top scaling rung. (3) **Scaling rungs** at 1, 2, 4, 8, 16, and 28 training sets (nested, in release order) plus two composition probes, all with the fixed recipe and a shared step cap. (4) **Ablations**: A-notext (structured features only, no text embedding) and A-noUB (the three licensed-IP training sets removed), for the shift analysis of §7. (5) **Baselines**: random; an hour-zero rarity-color heuristic; and three asterisked post-release baselines (community site ratings at day ∼2, and two 17Lands-statistics pickers) that bound what cheap post-release information buys. (6) **Post-day-one rows**, frozen now and executed only after the zero-shot evaluation: a per-set ceiling trained on the frozen MSH snapshot with the stock per-set recipe, and F-full fine-tuned on the MSH training split with hyperparameters pinned from dev-trio trial runs.

5.5 Claims bands

To keep interpretation out of our own hands, three mutually exclusive claims paragraphs (for headline results below 45%, between 45 and 55%, and above 55%) were drafted before any test number existed and are carried in the paper source; the matching one is selected verbatim when the frozen evaluation reports (§6). In all bands the secondary claims are the same: the normalized-vs-ceiling score, the shape of the scaling curve, the licensed-IP shift analysis, and the skill-conditioning effect.

6 Results

Zero-shot transfer on the development trio. Table 1 reports zero-shot transfer on the development trio: F-dev, the recipe used for architecture and hyperparameter selection, reaches 54.3% agreement, trained on 149.4M picks from 28 sets and evaluated on three sets it has never seen a pick from. (The paper's actual headline result is the frozen MSH number reported below; every number in this paragraph is a dev-trio rehearsal by the protocol's own design, §5.) Broken out per set, DraftFM reaches 48.9% / 57.5% / 56.5% on BRO / TMT / SOS against per-set ceilings of 65.6% / 71.0% / 70.2%, recovering 74.3% / 81.3% / 80.2% of those ceilings by the pre-registered normalized score (dev-mean 78.6%; aligning zero-shot and ceiling to identical held-out picks, §5). The simpler unaligned ratio over the full slice – 74.5% / 81.0% / 80.5%, dev-mean 78.7% – corroborates this on a much larger sample (the aligned subset is capped by the ceiling model's own held-out split) but is not itself the pre-registered metric. Excluding forced picks lowers the three numbers to 45.2% / 54.1% / 53.2%. The deployment-mode skill bucket selected on dev is the 0.66 win-rate bucket.

Reading the BRO row. BRO is the holdout on which Bertram et al. [3] report 55.4%: the strongest published unseen-set number, achieved with card features, learned image embeddings, *and* 16 usage statistics computed from human play after the set's release, on an all-users (not expert-filtered) pick population. Our 48.9% (expert slice; 48.4% all-users, the population-matched comparator) is below it. The comparison the BRO row is designed for is the day-one setting, where usage statistics do not exist: against the same paper's features-only representation (33.6–35.6%, single-set training) and every day-one-feasible baseline in Table 2, multi-set training on public card features closes most of the distance toward the usage-statistics-inclusive number while using strictly release-day information. BRO is also the trio's hardest set for transfer: its packs mix in a 63-card retro-artifact bonus sheet, examined directly in §7. The gap between the BRO and TMT/SOS rows – and why TMT, the trio's licensed-IP dev set, nonetheless transfers as well as SOS – is analysed there too; Figure 4 shows a real P1P1 bomb from TMT scored the same way. For sets in the training corpus but without a dedicated per-set model, the deployed system instead falls back to F-full's scoring; Figure 5 shows what that looks like today for LTR, LCI, and DMU – a deployment illustration, not an additional zero-shot data point.

Baselines. Table 2: random picking scores 23.2%, and an hour-zero heuristic (prefer lower rarity for playables, follow the pool's colors) reaches 42.8%: just below GPT-4o's 43% [2] and close to

Table 1: Zero-shot expert-slice top-1 agreement (%), deployment mode, forced picks included; 95% cluster-bootstrap CIs over drafts in parentheses. F-full's dev cells are never quoted (it trains on those sets). Each cell's source run is recorded as a comment in the generated table file.

Model	BRO	TMT	SOS	Dev mean	MSH
DraftFM (F-dev, zero-shot)	48.9 (48.8, 49.0)	57.5 (57.3, 57.7)	56.5 (56.4, 56.6)	54.3	TBD
DraftFM (A-noctx, selected dev recipe)	50.0 (49.9, 50.1)	57.7 (57.5, 57.9)	56.6 (56.5, 56.7)	54.8	–
DraftFM (F-full, zero-shot)	–	–	–	–	TBD
Per-set ceiling	65.6	71.0	70.2 (69.8, 70.6)	68.9	TBD
F-dev / ceiling (unaligned)	0.745	0.810	0.805	0.787	TBD

Figure 3: Steel Seraph, BRO – the dev trio's hardest transfer set (§7). Real P1P1 expected value 9.59, the 99.4th percentile of BRO's card pool by the deployed per-set model's own ranking; the next two candidates in the pack, Skystrike Officer and Tyrant of Kher Ridges, rank at the 99.0th and 98.7th. Percentile, not a head-to-head split: Arena hides the pack between picks in this mode (§4.2). Never MSH.

the expert-tuned 44.5% of Ward et al. [10], a useful caution about how much of drafting is surface regularity. DraftFM's zero-shot numbers, on each of the three dev sets, clear every day-one-feasible baseline in the table (Random and the rarity heuristic are measured on SOS as a rehearsal stand-in, not per-set). The three asterisked post-release baselines bound what cheap human data buys within days of release. Their MSH rows are filled in once the frozen evaluation runs.

The frozen MSH evaluation. The headline: F-full expert-slice deployment-mode top-1 on MSH is [PENDING: F-full MSH top-1]% (CI in Table 1). Snapshot gates, identifiers, and hashes are reported in the companion implementation reference (Appendix B points there). [PENDING: MSH secondary metrics: top-3, all-users top-1, normalized score vs the post-day-one MSH ceiling, log-loss, ECE, late-draft retention]

Scaling. Figure 6 and Table 3 report zero-shot accuracy as training grows from one set (NEO, 5.4M picks) to the full 28-set universe (149.4M picks), under the fixed recipe. Single-set training is the regime of Bertram et al. [3]'s features-only anchor; the top rung is F-dev. Two extra seeds at S1 and S4 give within-training top-1 spreads of 0.677–0.682 and 0.688–0.689 – training itself is not noise-dominated at either rung, checking in-distribution training stability and complementing the zero-shot dev-mean seed spread reported below. A log-linear fit of dev-mean top-1 on $\log_2$(training picks) across the nested ladder (S1 through F-dev, 5.4M–149.4M picks) tracks the six rungs closely – slope 1.0 percentage point per doubling, $R^2 = 0.98$, no rung off the fitted line by more than 0.4pp

Figure 4: April O'Neil, Hacktivist, TMT – the dev trio's licensed-IP set (§7), a comic-book bomb the text encoder never saw described in *Magic: The Gathering* rules-text style. Real P1P1 expected value 6.22, the 96.7th percentile of TMT's card pool. Never MSH.

Table 2: Baselines and published anchors. Upper block: our measured baselines, expert-slice deployment-mode top-1 (practice rows will be replaced by frozen MSH rows by the same pipeline). Lower block: published numbers, quoted from their papers on their own (general-player, not expert-filtered) pick populations; test sets, information regimes, and pick populations all differ from the upper block and from each other, so these situate rather than compare. * = uses post-release or otherwise non-day-one information.

Method	Information used	Test set	Top-1 agreement (%)
Random	none	SOS PremierDraft (rehearsal)	23.2 (23.2, 23.3)
Rarity–color heuristic	card features (hour 0)	SOS PremierDraft (rehearsal)	42.8 (42.7, 42.9)
Site-ratings heuristic (day ∼2)*	post-release statistics	MSH	TBD
ALSA-argmin*	post-release statistics	MSH	TBD
shrunk-GIH-argmax*	post-release statistics	MSH	TBD
DraftsimBot [10]	expert-tuned ratings	M19 (Draftsim)	44.5
GPT-4o zero-shot* [2]	LLM pretraining	NEO	43.0
Features-only, single-set [3]	card features	unseen (NEO-trained)	33.6–35.6
Features+meta+image, 13-set* [3]	post-release meta + images	BRO (unseen)	55.4

– but that same fit argues against, not for, reading a diminishing-returns point into the tested range: the per-doubling gain moves between 0.7 and 1.7pp/doubling non-monotonically (lowest across the S2–S8 middle of the ladder, highest at the S1–S2 start and the S16–F-dev end) rather than decaying toward the top rung. Six error-bar-free points on a fit whose own top rung is the model selected on this axis are not strong evidence of log-linearity; the honest reading is narrower than it looks: no diminishing returns are visible over the two orders of magnitude tested, not that returns have been shown not to saturate beyond it. Re-scoring those same extra-seed checkpoints on the zero-shot dev-mean the curve plots – the metric on the axis, not within-training top-1 – gives per-seed spreads of 0.65pp at S1 and 0.42pp at S4, at most about a quarter of the 2.82pp gain from S1 to S4, so the rung-to-rung climb is not a seed artifact. These are three-seed spreads at two rungs that also fold in an early-stop-step difference (the two added seeds stopped at half the reference seed's step), so they bound the curve's seed noise rather than fully characterize it, and stand at S1/S4 as a proxy for the top rung. [PENDING: whether set count or pick count better predicts the curve (S2b/S4b probes)]

LTR: EV 6.33, pct 99.4

LCI: EV 6.10, pct 99.5

DMU: EV 5.78, pct 99.2

Figure 5: Three sets with no dedicated per-set model – Lord of the Rings, The Lost Caverns of Ixalan, Dominaria United – all in F-full's 31-set training corpus. This illustrates F-full's fallback scoring path for training-corpus sets without a dedicated per-set model – the same path a newly released set would use before a per-set model exists for it. Real EVs and percentiles, computed the same way as Figures 3 and 4 – but because F-full trained on these three sets, this illustrates the production fallback path, not a zero-shot-accuracy claim; that claim is Table 1's dev-trio and MSH rows only.

Table 3: Scaling ladder. Rung labels are frozen protocol identifiers, not set counts (*S27* names the top rung's run, which trains on 28 sets – a naming artifact from an earlier corpus size, kept as-is rather than renamed after the fact); set and pick counts in the table are the real values from run records. Dev-mean is the model-selection metric of §5.

Rung	Sets	#sets	Train picks (M)	Best step	Dev-mean top-1 (%)
S1	NEO	1	5.4	4000	48.9
S2	S1 + DSK	2	13.3	18000	50.9
S2b	MOM, TDM (probe)	2	14.5	8000	50.2
S4	S2 + DMU, FIN	4	27.8	24000	51.7
S4b	STX, SNC, OTJ, TLA (probe)	4	23.5	20000	51.0
S8	S4 + STX, MOM, BLB, TLA	8	52.9	18000	52.4
S16	S8 + AFR, SNC, ONE, LTR, WOE, MKM, OTJ, EOE	16	99.9	34000	53.3
S27	full F-dev universe	28	149.4	34000	54.3

7 Analysis

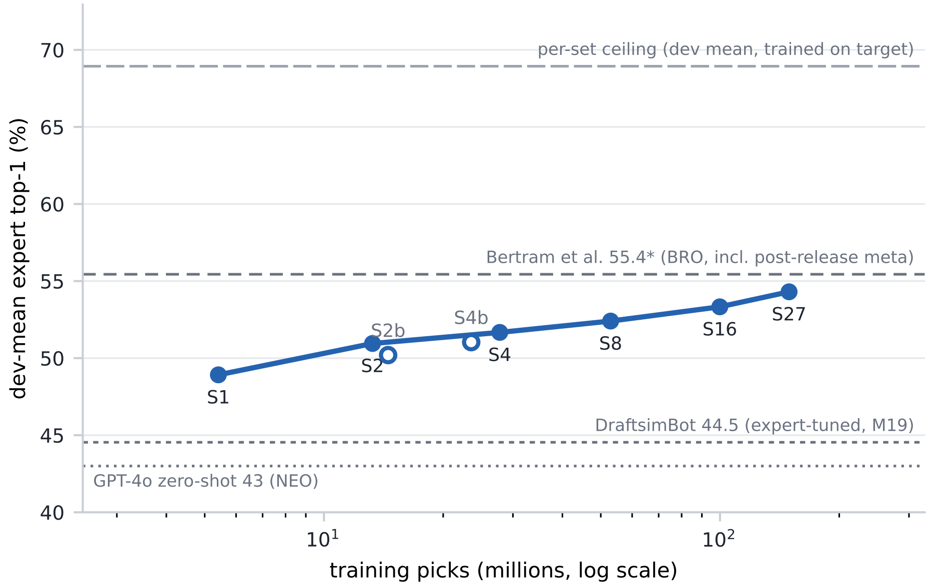

Figure 6: Zero-shot dev-trio-mean expert top-1 vs training picks (log scale). Filled points: the nested scaling ladder; open points: set-composition probes. Horizontal rules: published anchors and the per-set ceiling mean. Generated from run records by `figures/scaling_curve.py`; rungs appear as their frozen evaluations land.

Seed variance on the top-scoring ablation. Two additional seeds of A-noctx (the recipe F-full uses, §4.2) land within-training top-1 at 0.680–0.684, a 0.5pp spread around the seed reported in the table – A-noctx's own accuracy is not a single lucky initialization, though the close margins between it and the neighboring recipe-search variants (54.2–54.8 dev-mean, Table 4) mean the full ranking has not been seed-checked.

The flexibility-clause extras. The protocol's flexibility clause (§5) admits VOW QuickDraft and the Powered Cube into F-full's training data only if a dev-trio ablation shows a gain over A-noctx. A-noctx plus VOW QuickDraft (same recipe and holdout, one shard added) scores 48.9/57.6/56.0, dev-mean 54.2 – lower than A-noctx's 50.0/57.7/56.6/54.8 on all three dev sets, by a paired-difference CI that excludes zero on every set (−1.1pp BRO, CI −1.2 to −1.1; −0.1pp TMT, CI −0.26 to −0.03; −0.6pp SOS, CI −0.62 to −0.52). The clause's own rule therefore excludes it: F-full's frozen recipe (§4.2) does not train on VOW QuickDraft. The Powered Cube half of the clause was not tested at all – bulk per-pick data for it is not available through the public 17Lands pipeline this paper otherwise relies on exclusively, and acquiring it by another route was out of scope. Both extras are therefore absent from F-full's training data: one by a pre-registered failed ablation, the other by an unmet data precondition, and the frozen recipe is exactly the 31-set corpus described in §3.

The licensed-IP shift. MSH's names and flavor come from a comic universe, not from *Magic: The Gathering*. Its text lives in the sentence encoder's pretraining as a different genre. This is the distribution shift most likely to break a text-dependent model on the day it matters, and it is why the protocol makes a licensed-IP set (TMT) a development set: the rehearsal happens where it can still inform design. One mitigation is built in: card names are masked out of the

Table 4: Zero-shot expert top-1 (%), point estimates (per-cell 95% CI half-widths are $\sim \pm$ 0.1–0.2pp on the dev sets, Table 1; paired-difference CIs accompany the deltas in the text). A-noctx, A-proportional, and A-topfilter are recipe-search variants against the Full baseline, run before the deployed architecture was fixed; A-notext and A-noUB are the pre-registered ablations (§5) analysed below. A-notext drops the 384-d text embedding from the Full recipe. A-noUB removes the three licensed-IP training sets from A-noctx (the winning architecture, §4.2), so its clean comparator is the A-noctx row, not Full. The (variant $\times$ set) grid supplies the dev-side numbers for the licensed-IP text penalty (per-set deltas below), computed before MSH is seen.

Variant	BRO	TMT	SOS	Dev mean	MSH
Full (F-dev recipe)	48.9	57.5	56.5	54.3	TBD
A-notext	45.8	55.9	53.1	51.6	TBD
A-noctx (winning)	50.0	57.7	56.6	54.8	TBD
A-proportional	50.2	56.8	55.7	54.2	TBD
A-topfilter	49.8	57.3	56.5	54.5	TBD
A-noUB	49.2	57.5	54.9	53.9	TBD

text before embedding (§4.1), so the encoder sees mechanics, not franchises. A second design choice does not reduce the shift but makes it measurable: the ablation grid of Table 4 separates what the text embedding contributes on in-universe sets (SOS, BRO) from what it contributes on licensed sets (TMT, MSH). Zero-shot, the licensed dev set is not the hard one: TMT reaches 57.5% against SOS's 56.5% (and 81.0% vs 80.5% of the respective ceilings), suggesting the masking plus structured features carry the load. Whether that holds because text adds little on licensed sets or because TMT is simply an easier format is what the grid decides: dropping text costs TMT only 1.6pp (CI 1.5–1.7pp) against its Full comparator, less than BRO's 3.1pp (CI 3.0–3.2pp) or SOS's 3.4pp (CI 3.3–3.5pp) – text contributes least on the licensed-IP set, favoring the masking-plus-structured-features explanation over TMT simply being an easier format. The UB-penalty (removing the three licensed-IP training sets from A-noctx) tells a consistent story: 0.8pp on BRO (CI 0.7–0.9pp), 0.2pp on TMT (CI 0.09–0.33pp), 1.7pp on SOS (CI 1.6–1.7pp) – losing licensed-IP training data costs the licensed-IP dev set least, not most, and even TMT's small penalty is a real, paired-CI-supported effect, not noise dressed up as a trend. If the test set is harder than the trio average, the pre-registered framing stands: a licensed-IP test makes the zero-shot claim more credible, not less.

Why BRO transfers worst. BRO sits 8.1pp below the mean of TMT and SOS and only 74.5% of its ceiling. Of the three candidate explanations, the bonus sheet looked, at first pass, like the most direct: packs offering at least one of the 63-card retro-artifact bonus sheet score 40.6% zero-shot, packs that don't score 55.5% – a 14.8pp raw gap, on 44% of BRO's picks. That gap is mostly a different artifact in disguise: BRO's booster carries exactly one bonus-sheet card, so a pack can only "have the bonus card" before it has been picked away, and *has_bonus* falls from 100% at pick 1 to 2.9% at pick 15 (BRO's last, forced pick) by construction – bonus-present packs average pick 4.1, bonus-absent packs average pick 9.3, and early picks are mechanically harder regardless of what the pack contains. Stratifying by pick number before comparing (Figure 7) collapses the gap to 3.1pp (CI 2.8–3.4), roughly 16% of BRO's shortfall against the trio, not something comparable in size to it. The per-(pack, pick) curve is the same family of artifact playing out differently: the raw BRO-vs-TMT/SOS gap widens through the pack, but BRO's booster carries one more card than TMT/SOS's (a real extra forced pick, itself a consequence of the bonus sheet enlarging the pool), so late raw picks are not apples-to-apples across sets; correcting for the random floor at each pick

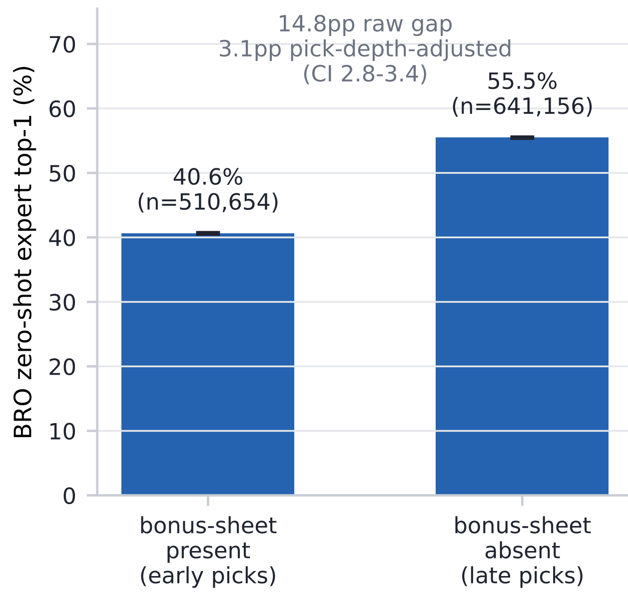

Figure 7: BRO zero-shot expert top-1, split by whether the pack offered at least one of the 63 BRR retro-artifact bonus-sheet cards (44% of BRO's picks do). The raw 14.8pp gap is mostly a pick-depth confound (labeled above the bars): *has_bonus* tracks early-vs-late pick number by construction, and stratifying by pick number before comparing collapses the gap to 3.1pp (95% CI). Error bars on the raw bars are 95% cluster-bootstrap CIs over drafts. Generated from cached zero-shot predictions by `figures/bro_diagnostics.py`; never MSH.

reverses the trend. Artifact-heavy color-signal weakening and 2022-era recency remain plausible but not separately isolated by this grid – and, with the bonus-sheet slice's real contribution now measured at 16% rather than assumed to explain most of the shortfall, more of BRO's gap is left for them (and for factors this grid does not cover) to account for.

Skill conditioning. Conditioning is a scorer-side input (§4.2), so one model serves any audience. In human-model mode (§5) the model scores 48.4% / 57.8% / 56.8% on BRO / TMT / SOS: within a point of the expert-slice deployment numbers, consistent with skill moving picks less than card quality does. The pre-registered effect size (deployment-mode accuracy as a function of the conditioning bucket, and the gap between conditioning on the top bucket versus the drafter's own) is [PENDING: skill-conditioning effect on MSH].

Calibration. Zero-shot confidence falls well short of within-set calibration: top-label ECE is 0.102 / 0.071 / 0.093 on the dev trio, against 0.014 for the within-set SOS ceiling, roughly a five- to seven-and-a-half-fold gap. The protocol permits one calibration decision: a temperature fitted on dev only and frozen into the battery, never fitted on MSH. That temperature is 1.28, chosen by minimizing mean dev-trio log-loss over a grid (not ECE directly, since log-loss is the smooth, proper scoring rule this protocol's other dev decisions already use); it cuts the dev-trio mean ECE from 0.089 at the unscaled T=1 to 0.025, most of the five-to-seven-and-a-half-fold gap above. This temperature is fit on F-dev's logits (the with-tower recipe used for dev-side selection, §4.2) and applied unchanged to F-full (the tower-free recipe that scores MSH) at test time, since F-full has no held-out set of its own to fit a temperature against without touching MSH; MSH ECE, at this

Figure 8: The low-conviction end of the model's output: a real thirteen-card SOS pack (empty pool) where Sundering Archaic and Flow State score a near-exact tie (EV 1.41 vs. 1.41, dominance split 50/50) – the third candidate, Pursue the Past, trails at 27% dominance against the pair. Ties this close are exactly the regime the top-label ECE below is measuring: the model's stated confidence and its actual reliability at low conviction. Computed by the deployed per-set model, never MSH.

same cross-architecture frozen temperature, is [PENDING: MSH ECE].

Late-draft retention. Later picks are where a set-agnostic model could plausibly collapse to rate-the-card, since pool signal has taken over from the card alone. The pre-registered statistic, late-draft retention (§5), is 81.8% / 87.8% / 89.2% on BRO / TMT / SOS (dev-mean 86.3%) – higher, not lower, than the corresponding all-pick ratios (74.5% / 81.0% / 80.5%, §6). That raw ratio is not, on its own, evidence that the pool tower is adding signal late: picks 8+ carry roughly twice the forced-pick rate of the draft as a whole (forced picks score 1.0 by construction on both sides of the ratio) and a substantially smaller mean pack size (4.0–4.5 cards vs. 7.3–8.0 overall), and restricting to genuine choices (pack size ≥ 4, excluding the trivial tail) puts late-pick top-1 *below* the all-pick figure on all three dev sets, the opposite direction from the retention ratio. The claim the data does support is narrower: the model does not collapse to chance late in the draft. The per-(pack, pick) curves against the per-cell random floor show this directly – excluding forced picks, zero-shot top-1 never comes closer than 20.5 (BRO) / 32.5 (TMT) / 28.4 (SOS) percentage points to chance at any point in the draft, including its latest picks. [PENDING: late-draft retention on MSH]

Acknowledgments

This work would not exist without 17Lands, whose public datasets [1] are the training and evaluation corpus, and whose data-sharing policy makes reproducible drafting research possible. Card metadata is provided by Scryfall [9]. *Magic: The Gathering* is the property of Wizards of the Coast; this is unofficial research, unaffiliated with and unendorsed by Wizards of the Coast, 17Lands, or Scryfall.

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A Limitations

The target is agreement, not optimality. Top-1 agreement with expert humans measures imitation. Experts disagree with each other (the within-set ceiling is ∼70%, not 100%), and agreeing with them is not the same as maximising win rate. A model could in principle draft better than its agreement suggests, or worse. Win-rate evaluation of zero-shot drafters is future work.

Population. 17Lands users are self-selected: they run a tracker, they draft on Arena, and the expert slice further conditions on volume and success. All numbers are relative to this population; paper drafting, other platforms, and casual play are out of scope.

One evaluation round. Pre-registration buys credibility at the cost of adaptivity: the frozen battery cannot react to surprises in the test snapshot, and a single set is a sample of one from the distribution of future sets. The dev-trio spread and the composition probes bound, but do not eliminate, set-to-set variance in what the headline number would be on the *next* set.

Text and language. The text encoder is English-only and frozen. Rules text is embedded as prose, so mechanics whose meaning is mostly numeric or positional may be under-represented. Name masking removes the most direct pretraining leak for licensed sets, but flavor vocabulary in the remaining text still differs by universe (§7).

Numerics. Training runs on an accelerator backend with documented silent-error classes; we gate on CPU parity and skip non-finite steps (§4.3), which guards the classes we know about.

B Reproducibility

Every result in this paper traces to a run id in an append-only ledger, and every table cell and prose number is emitted by a script from that ledger and cached evaluation outputs – no result number here is typed by hand. The ledger schema, frozen identifiers (protocol tag, manifest hashes), release contents, and script-level pointers for reproducing any specific number are in the companion implementation reference, not repeated here.

C Licensing and attribution

Draft data. Draft data is from the 17Lands public datasets [1], used under CC BY 4.0 and per the project's usage guidelines (bulk S3 downloads; no API scraping).

Card metadata. Card metadata is from Scryfall [9], used per its API guidelines; Scryfall neither produces nor endorses this work.

Card images. Figures 1, 3, 4, 5, and 8 reproduce official Scryfall card scans [9], fetched directly from Scryfall's own hosted image URLs (fetch script and source manifest released with the companion reference). Scryfall's stated policy on card images ("Use of Scryfall Data and Images," `https://scryfall.com/docs/api`) permits this use, subject to rules we follow throughout: every image is shown full and unmodified, with no cropping, distorting, blurring, sharpening, desaturating, color-shifting, or watermarking of the card art itself. Any score or label in a figure sits beside the image, never on top of it. Because the full card image is always used (never an art crop), the copyright and artist line Scryfall prints at the bottom of the card remains visible in every figure, and nothing here implies that Wizards of the Coast produced or endorses this work. The cards illustrated are drawn from BRO, SOS, TMT, LTR, LCI, and DMU – all public, released sets (§3) – never from MSH, which remains untouched until the frozen evaluation of §5.